



LUND
UNIVERSITY

Written Examination
Topology MATM16
May 25, 2018
Time: 08.00–13.00

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Use the papers provided by the department. Write clearly with short and concise motivations. Illustrate with a figure when necessary. Please write on the last page one or several suggested times for oral.

1. Let $\mathcal{B}$ be the family of sets of the form $(a, b]$ where $a, b \in \mathbb{R}$ and $a < b$.
 - (a) Prove that $\mathcal{B}$ is a basis for a topology $\mathcal{T}$ on $\mathbb{R}$ which is strictly larger than the usual topology on $\mathbb{R}$.
 - (b) Prove that each interval $(a, b]$ is $\mathcal{T}$ -closed.
 - (c) Is $(\mathbb{R}, \mathcal{T})$ connected?
 - (d) Identify the connected components (maximal connected sets) of $(\mathbb{R}, \mathcal{T})$.
2.
 - (a) Define what is meant by a normal topological space.
 - (b) Let X be a normal topological space, F is a closed subset of X , and V an open subset of X such that $F \subset V$. Prove that there is an open subset U of X such that $F \subset U \subset \bar{U} \subset V$.
3. Let X be a topological space with a given subbase $\mathcal{S}$ for the topology, and $f : X \rightarrow X$ a function. Decide whether or not the following statements are true in general. (Also supply your statements with proofs or counterexamples.)
 - (a) The set of products $U \times V$ where $U, V \in \mathcal{S}$ is a subbase for the product topology on $X \times X$.
 - (b) f is continuous if $f^{-1}(V)$ is open in X for each $V \in \mathcal{S}$.
 - (c) f is an open map if $f(U)$ is open in X for each $U \in \mathcal{S}$.Hint: For (c) you could consider $X = \{a, b, c\}$ and $\mathcal{S} = \{\{a, b\}, \{b, c\}\}$.
4. Let $X = C([0, 1])$ denote the space of all continuous real-valued functions on $[0, 1]$ with the metric of uniform convergence $d(f, g) = \sup\{|f(x) - g(x)|; 0 \leq x \leq 1\}$.
 - (a) Prove that, for each given $n \in \mathbb{N}$, the set

$$E_n = \{f \in X; |f(x) - f(1/2)| \leq n|x - 1/2|, 0 \leq x \leq 1\}$$

is closed and nowhere dense in X .

- (b) Consider the subset $D \subset X$ of all functions in X that are differentiable at $x = 1/2$. Prove that the complement $X \setminus D$ is dense in X .

Please, turn over!

5. A metric space (X, d) is called *topologically complete* if it is possible to define an equivalent metric ρ on X such that (X, ρ) is complete.

The goal of the following problem is to show that if X is an open subspace of $\mathbb{R}^n$ (with the usual metric $d(x, y) = |x - y|$) then X is topologically complete. To exclude trivial cases, we will assume that X is nonempty and that X is not all of $\mathbb{R}^n$.

(a) Prove that (X, d) is not complete.

(b) Let $F = \mathbb{R}^n \setminus X$ and define $d(x, F) = \inf\{d(x, y); y \in F\}$. Prove that $x \mapsto d(x, F)$ is a strictly positive and Lipschitz continuous function on (X, d) :

$$|d(x, F) - d(x', F)| \leq d(x, x').$$

(c) Write $f(x) = 1/d(x, F)$ for $x \in X$ and put $\rho(x, y) = d(x, y) + \min\{1, |f(x) - f(y)|\}$. Prove that ρ is a metric on X and that d and ρ are equivalent on X , i.e., $d(x_n, x) \rightarrow 0$ where $x \in X$ if and only if $\rho(x_n, x) \rightarrow 0$.

(d) Prove that (X, ρ) is complete.

(e) Prove that (X, ρ) is separable.

Note: A separable and topologically complete space is called a *Polish space*.